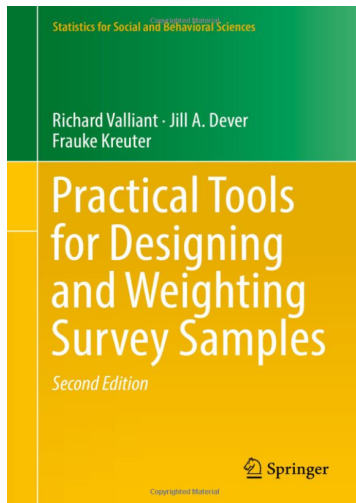


Practical Tools for Study Design and Inference

Introduction

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What is this class about? - 1

- It will be assumed that you have taken an introductory sampling class.
- At times, you may need to review materials from the applied sampling or sampling theory class to get the full grasp of the materials in this class.

What is this class about? - 2

- ① Single-stage surveys — sample size calculations (CV targets, power), different types of designs (*srs*, *stsr*s, *pps*)
- ② Math programming
- ③ Adjustments for sample losses
- ④ Multistage samples — 2- and 3-stage designs, variance components, allocations to stages
- ⑤ Area Samples — Census geographic units, composite MOS
- ⑥ Steps in weighting — base weights, adjustments for unknown nonresponse, use of auxiliary data
- ⑦ Calibration — poststratification, raking, GREG, weight variability
- ⑧ Weighting adjustment approaches for nonprobability sampling
- ⑨ Sample size calculation
 - Estimate precision
 - Statistical power for hypothesis testing

Why should you care?

- ① You will be considered as the survey guru who people will look to when it comes to designing surveys or analyzing survey data.
- ② What are the types of questions your clients will try to answer with the survey data?
- ③ What are the types of surveys your clients will use?
- ④ What are the types of sampling your clients will use?
- ⑤ What are the types of questions your clients will have for you?

How does this class work?

- Me: Deliver course materials through lectures and lab codes
- You:
 - Prepare each class by **completing** assigned reading and working on the code
 - Complete homework assignments (individual) and a project (group) on time
- **Practical** tools through programming
- Study design and inference through **math**
- Grades will consider both individual work and group work

Let's go over the syllabus.